

Undead Arcade

Game Design Report

Phaser 3.90 | Vite | 1280x720 | Supabase scores | Package: undead-arcade

3 player hearts	7 enemy types	2 abilities	2+ arena levels
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1. What the game is about

Undead Arcade is a classic arcade-style 2D top-down zombie survival shooter. You play as a customizable soldier in a scrolling apocalyptic arena. Zombies enter from the borders; you shoot them, collect coins, buy shop upgrades, pick up rare abilities, defeat a boss, and walk through an exit door into the next level. There is no permanent win screen - the run ends when HP reaches zero. On death, you can save survival time and progress to a Supabase leaderboard.

Goal

Survive as long as possible. Clear arenas, beat bosses, advance levels, and stack coins/upgrades between arenas.

Lose condition

Reach 0 hearts -> death overlay -> Save score / Restart / Main Menu (Supabase run_scores table).

Level timeline

	Level 1	Level 2+
Map	Map.png	map2.png
Spawn stop (clear window)	60 seconds	120 seconds
Boss unlock	90 seconds	150 seconds
Enemy cast	Walkers + bone throwers	Walkers, crawlers, exploders, acid spitters
Boss	Charger (Boss1)	Acid Broodmother (Boss2)
Music	Level 1 loop -> bossbattle	Level 2 loop -> bossbattle2

2. Characters

Player - Soldier

Built from a headless body spritesheet (Mainbody.png) with an optional webcam or uploaded face baked into a retro pixel head and attached at the neck socket. Aim follows the pointer; movement uses WASD.

Stat	Value
Max HP	3 hearts
Move speed	155
Bullet damage / speed	1 / 520
Base fire rate	420 ms
Magazine / reload	8 rounds / 1100 ms (press R)
I-frames after hit	5000 ms
Knockback	220

Enemies

Type	HP	Speed	Behavior
Walker (Zombie1)	3	45-75	Predictive chase + pack separation
Bone thrower (Zombie2)	2	28-42	Keeps range ~320; fires every 1.7s
Exploder	7	120-150	Rushes; detonates near player or on death
Acid spitter	3	32-48	Long-range spit + acid puddles
Crawler	2	155-190	Fast ground rush; holds close range
Boss 1 - Charger	70	115-130	Chase, 5-bone fan, telegraph charge
Boss 2 - Broodmother	95	95-115	Acid, birth crawlers, burrow, throw exploder

Spawn pacing: continuous timed spawns (not numbered waves). Pool cap 28. Level 2 unlocks spitters (~18s), crawlers (~8s), and rarer exploders (max 1 at a time).

3. Abilities

Non-boss kills have a 5% chance to drop an ability pickup (50% in tester mode). Shockwave art exists in assets but was removed from gameplay.

Overdrive (7 seconds)

Fire rate x0.55 (floor 120 ms), infinite ammo, cancels reload. Orange tint + voice SFX.

Shield (1 charge)

Blue ring. Next hit shows "BLOCKED", no HP loss, still grants 5 seconds of i-frames.

4. Mechanics

Controls

- WASD - move (arrow keys disabled to avoid confusion)
- Click / hold / Space - shoot
- R - reload
- P - shop | Esc - pause

Economy

- 1 coin per normal kill
- Boss kill -> 20-coin splash + exit door
- Coin magnet radius 78
- Arena clear -> +1 heart (max 3)

Shop (press P)

Item	Cost	Effect
Heart	10	+1 HP (max 3)
Fire rate	12 / 25 / 42	x0.88 per level (floor 180 ms)
Mag size	10 / 20 / 36	Mag = 8 + 2x level (up to 14)
Reload	14 / 28	x0.82 per level (floor 450 ms)

Face customization pipeline

Menu webcam snapshot or upload -> 512x512 crop -> POST /api/bake-face (Gemini / OpenAI) green-screen head -> chroma key -> localStorage -> composited onto soldier sheet as a retro 16-bit style head.

Level loop

Timed spawns -> spawn stop -> clear remaining enemies -> optional heart reward -> boss warning (4s alarm) -> boss fight -> door -> LEVEL CLEAR -> next level. Coins/upgrades persist; HP/ammo refill.

5. Systems and key functions

Module	Role
src/main.js	Phaser boot 1280x720 MenuScene -> GameScene
src/scenes/MenuScene.js	Face capture, bake, START, hidden tester unlock
src/scenes/GameScene.js	Core loop: spawn, combat, shop, levels, audio, death
src/zombies.js	Enemy types, AI, bosses, sheet baking
src/soldierAnim.js	Soldier sheet + head attach
src/deathOverlay.js + scores.js	Death UI + Supabase run persistence
server/bakeFace.js	Vite middleware /api/bake-face
supabase/.../run_scores.sql	Leaderboard table + anon RLS

Important functions

Area	Functions
Spawning	handleZombieSpawns, spawnIntervalForElapsed, spawnZombie
AI	updateZombies, updateBoss, updateBroodmotherBoss
Combat	handleMovement, handleShooting, fireBullet, hurtPlayer, killPlayer
Abilities	activateOverdrive, activateShield, collectAbility
Economy	spawnCoin, spawnCoinSplash, openShop, recalcWeaponStats
Face	generatePixelHead, buildSoldierAnim, createBakeFaceMiddleware
Scores	saveRunScore, fetchTopScores

Hidden tester mode: click the title 5 times on the menu to unlock TEST: BOSS / BOSS 2 / GOD / LEVEL 2 (extra coins; ability drops more often).

6. Prompts used to build the game

The game was built iteratively in Cursor from natural-language requests, plus image and music generation prompts for Sorceress / Gemini. API keys are omitted.

Original concept (proposal)

"classic arcade 2D zombie shooting game with customizable character to be the player. every now and then the zombie will drop abilities or item which can be used."

From project proposal notes | Aug 7, 2026

Kickoff build prompt

"i want to build a top down shooting game. you're a soldier that starts in the middle and can shoot bullets at zombie that come in from the sides. I want to use phaser for this as the tech. Let's start off with the basics and just implement the soldier, movement, and shooting"

First implementation message | Aug 9, 2026

Feature prompts that shaped the game

Theme	Your request (summary)
Face custom	Pixelated face on soldier; camera snapshot + upload in menu
Sprites	Upload headless soldier body; left/right walk animation
World	Scrolling custom map; zombies from borders; scale difficulty
Combat	3 hearts; R to reload; 5s i-frames; muzzle spark
Boss L1	Stop spawns -> boss; charge attack; bone fan; warning + alarm
Economy	Coin drops; shop (P); arena clear -> +1 heart; boss 20-coin jackpot
Abilities	~1/10 drop idea -> Overdrive + Shield (Shockwave removed)
Level 2	New map/music; exploder, acid spitter, crawler; longer clear window
Boss L2	Female broodmother: acid, spawn crawlers, burrow, throw exploder
Polish	Title Undead Arcade; menu video; WASD-only; death save to Supabase

Asset generation prompts (examples)

Soldier body (Sorceress)

Create a pixel-art character spritesheet for a top-down zombie shooter.

Subject: Headless soldier body only; empty circular head socket.

Layout: 2 rows x 4 columns - Row1 RIGHT walk, Row2 LEFT walk.

Style: olive drab military, gun in shooting position, transparent PNG.

Exploder enemy

Create a character spritesheet for a top-down zombie shooter exploding enemy.

Subject: FAT bloated exploding zombie. Huge round belly; glowing orange-red cracks.

Layout: 2x4 - walk RIGHT / walk LEFT. Transparent. Match existing zombie art.

Acid spitter

Create a character spritesheet for a top-down zombie shooter acid spitter enemy.

Subject: Thin hunched zombie; swollen throat sacs, acid-stained clothes.

Layout: 2x4 left/right walk. Acid green accents. Transparent PNG.

Also generated with prompts: Boss1 / Boss2 sprites, crawler sheet, acid goo VFX, Level 1-2 maps, Level/Boss/Menu BGM, coin pickup SFX, Overdrive voice line, exploder timer beep, bosswarning alarm.

7. Quick reference

Metric	Value
Level 1 clear / boss	1:00 / 1:30
Level 2 clear / boss	2:00 / 2:30
Ability drop rate	5% (50% in tester mode)
Boss coin jackpot	20 coins
Player HP / speed / fire / mag / reload	3 / 155 / 420 ms / 8 / 1100 ms
Boss1 / Boss2 HP	70 / 95

Stats sourced from GameScene.js and zombies.js. Report generated Aug 12, 2026. Project path: Desktop/Zombie Shooter.